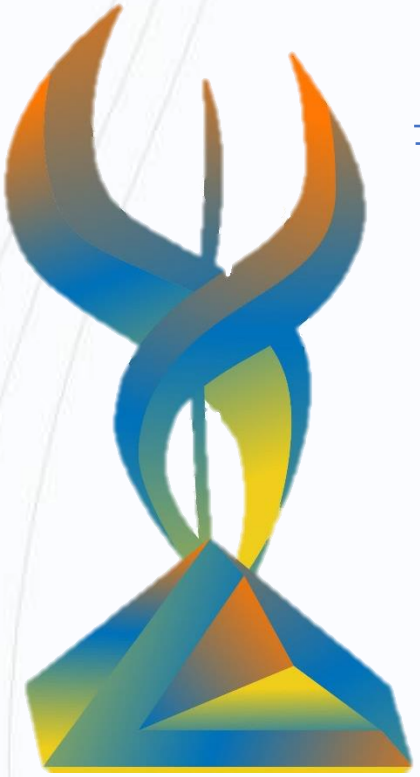
Republic of the Philippines
Bicol University
POLANGUI CAMPUS
Polangui, Albay

P2: FRONTEND DEVELOPMENT BASED ON PROJECT DESIGN



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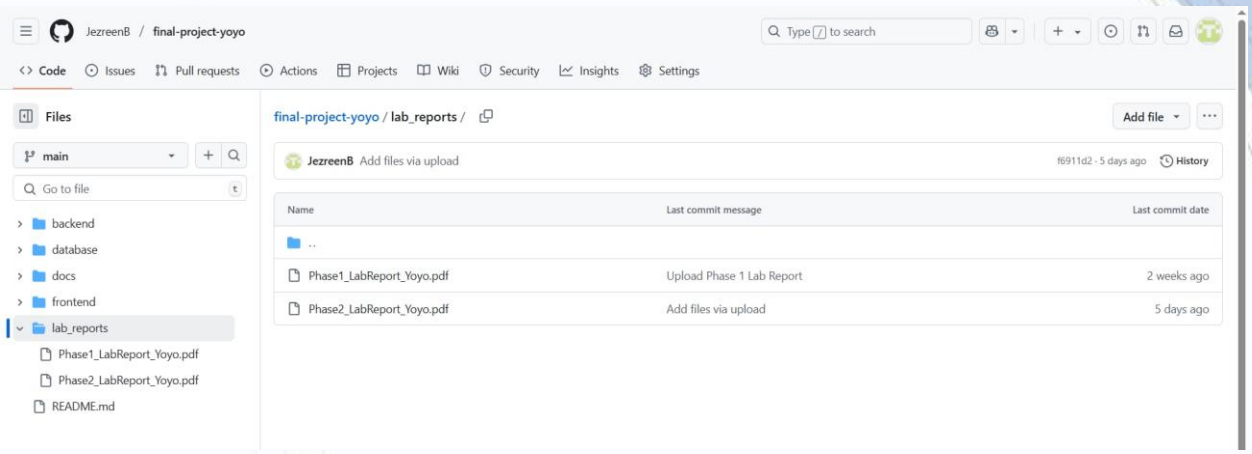
INITIAL LAB REPORT FOR PHASE 2 UPLOADED IN GITHUB

Content: All Working frontend pages with annotations



OVERVIEW OF BACKEND SETUP

1. **Technology Stack**

- Node.js with Express framework for the server.
- MongoDB with Mongoose ODM for database operations.
- JSON Web Tokens (JWT) for authentication.
- bcrypt for password hashing.
- Middleware for authentication and request parsing.

2. **Project Structure**

- server.js: Entry point of the backend server. Sets up Express app, connects to MongoDB, configures middleware (CORS, body-parser), and mounts route handlers.
- config/db.js: Handles MongoDB connection using Mongoose.
- routes/: Contains route definitions for different resources:
    - userRoutes.js: User registration and login.
    - productRoutes.js: Product listing and adding products.
    - reviewRoutes.js: Submitting and fetching product reviews.
    - cartRoutes.js: Managing shopping cart items.
    - orderRoutes.js: Placing orders and order management.
- controllers/: Contains controller logic for each resource, handling business logic and database interactions.
- models/: Mongoose schemas and models for User, Product, Review, CartItem, and Order.

- middleware/authenticateToken.js: Middleware to verify JWT tokens and protect routes.
- package.json: Lists dependencies and scripts for running the server.

3. Authentication

- Users register and login via /api/users routes.
- JWT tokens are issued on login and required for protected routes (adding products, managing cart, placing orders, submitting reviews).
- authenticateToken middleware verifies tokens on protected routes.

4. API Endpoints:

- /api/users: Register and login.
- /api/products: Get products, add product (admin only).
- /api/reviews: Submit and get reviews.
- /api/cart: Get, add, remove cart items.
- /api/orders: Place orders, get orders, cancel orders.

5. Security:

- Passwords are hashed with bcrypt.
- Role-based access control is implied (e.g., admin role for adding products).
- JWT tokens secure protected endpoints.

CODE SNIPPETS OF SERVER.JS, SAMPLE MODEL, AND ROUTES

SERVER.JS

backend > JS server.js > ...

```
require('dotenv').config();

const express = require('express');
const cors = require('cors');
const bodyParser = require('body-parser');
const connectDB = require('./config/db');
const userRoutes = require('./routes/userRoutes');

const app = express();
const PORT = process.env.PORT || 3001;

connectDB();

app.use(cors());
app.use(bodyParser.json());

app.use('/api/users', userRoutes);

const productRoutes = require('./routes/productRoutes');
const reviewRoutes = require('./routes/reviewRoutes');
const cartRoutes = require('./routes/cartRoutes');
const orderRoutes = require('./routes/orderRoutes');

app.use('/api/products', productRoutes);
app.use('/api/reviews', reviewRoutes);
app.use('/api/cart', cartRoutes);
app.use('/api/orders', orderRoutes);

app.listen(PORT, () => {
  console.log("Server running on http://localhost:" + PORT);
});

```

SAMPLE MODELS

backend > models > JS Product.js > ...

```
1   const mongoose = require('mongoose');
2
3   const productSchema = new mongoose.Schema({
4     name: { type: String, required: true },
5     category: String,
6     price: { type: Number, required: true },
7     image: String,
8   });
9
10  module.exports = mongoose.model('Product', productSchema);
11
```

backend > models > JS User.js > ...

```
1   const mongoose = require('mongoose');
2
3   const userSchema = new mongoose.Schema({
4     fullName: { type: String, required: true },
5     address: { type: String, required: true },
6     email: { type: String, required: true, unique: true },
7     password: { type: String, required: true },
8     role: { type: String, enum: ['user', 'admin'], default: 'user' }
9   });
10
11  module.exports = mongoose.model('User', userSchema);
12
```

backend > models > JS Order.js > ...

```
1   const mongoose = require('mongoose');
2
3   const orderItemSchema = new mongoose.Schema({
4     productId: { type: mongoose.Schema.Types.ObjectId, ref: 'Product', required: true },
5     quantity: { type: Number, required: true },
6     price: { type: Number, required: true },
7   });
8
9   const orderSchema = new mongoose.Schema({
10    userId: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
11    orderNumber: { type: String, required: true, unique: true },
12    status: { type: String, required: true },
13    total: { type: Number, required: true },
14    deliveryAddress: { type: String, required: true },
15    paymentMethod: { type: String, required: true },
16    shippingMethod: { type: String, required: true },
17    orderDate: { type: Date, default: Date.now },
18    items: [orderItemSchema],
19  });
20
21  module.exports = mongoose.model('Order', orderSchema);
22
```

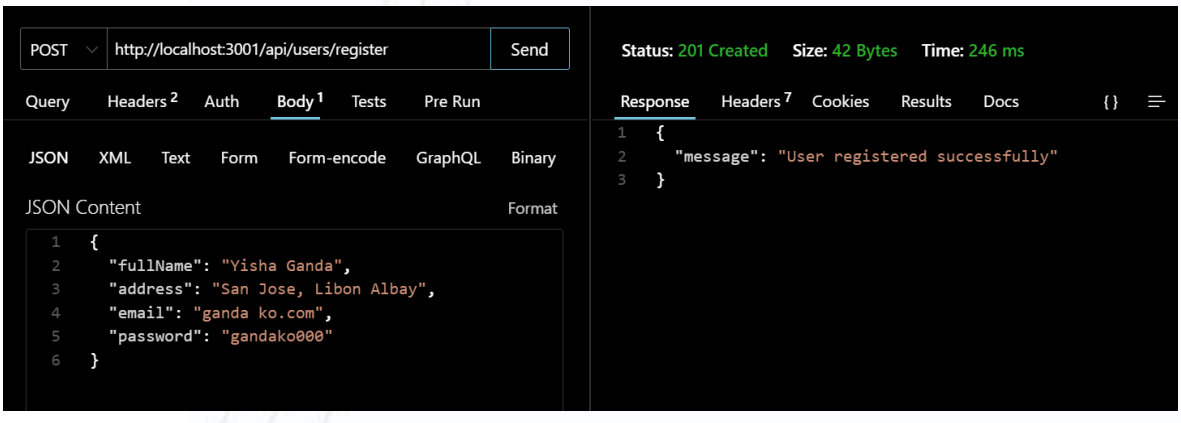
backend > models > JS Review.js > ...

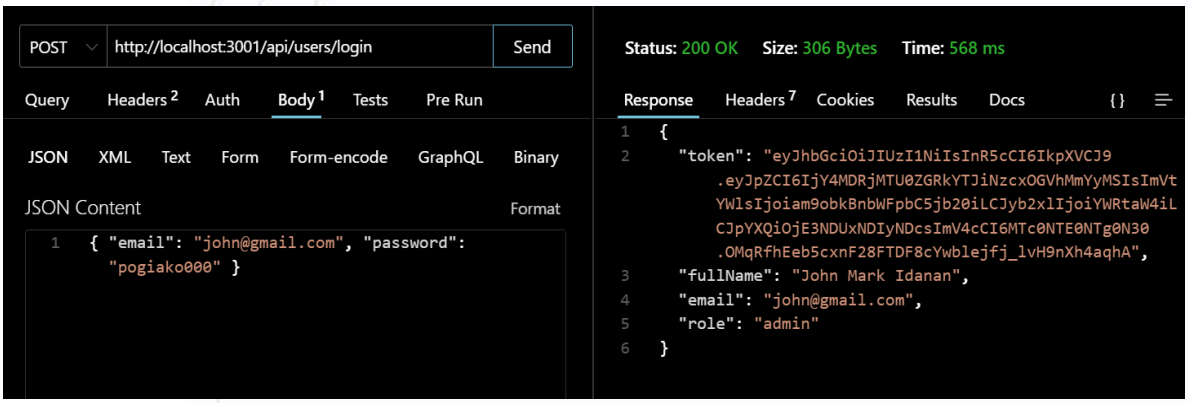
```js
const mongoose = require('mongoose');

const reviewSchema = new mongoose.Schema({
  productId: { type: mongoose.Schema.Types.ObjectId, ref: 'Product', required: true },
  userId: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  rating: { type: Number, required: true },
  text: String,
  date: { type: Date, default: Date.now },
});

module.exports = mongoose.model('Review', reviewSchema);

```

backend > models > JS CartItem.js > ...

```js
const mongoose = require('mongoose');

const cartItemSchema = new mongoose.Schema({
  userId: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  productId: { type: mongoose.Schema.Types.ObjectId, ref: 'Product', required: true },
  quantity: { type: Number, required: true },
});

module.exports = mongoose.model('CartItem', cartItemSchema);

```

API TESTING SCREENSHOTS





POST http://localhost:3001/api/products Send

Query Headers 3 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
{
"name": "NZXT H510 Elite",
"category": "PC Case",
"price": 89.99,
"image": "images/gc1.jpg"
}
```

Status: 201 Created Size: 27 Bytes Time: 71 ms

Response Headers 7 Cookies Results Docs { }

```
{
  "message": "Product added"
}
```

POST http://localhost:3001/api/orders Send

Query Headers 3 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
{
"deliveryAddress": "San Jose, Libon ALbay",
"paymentMethod": "Credit Card",
"shippingMethod": "Standard",
"items": [
{
"productId": "6804c242ddda2b7718ea2f25",
"price": 89.99,
"quantity": 2
}
]
}
```

Status: 201 Created Size: 52 Bytes Time: 5.97 s

Response Headers 7 Cookies Results Docs { }

```
{
  "message": "Order placed",
  "orderNumber": "NG-469740"
}
```

POST http://localhost:3001/api/cart Send

Query Headers 3 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

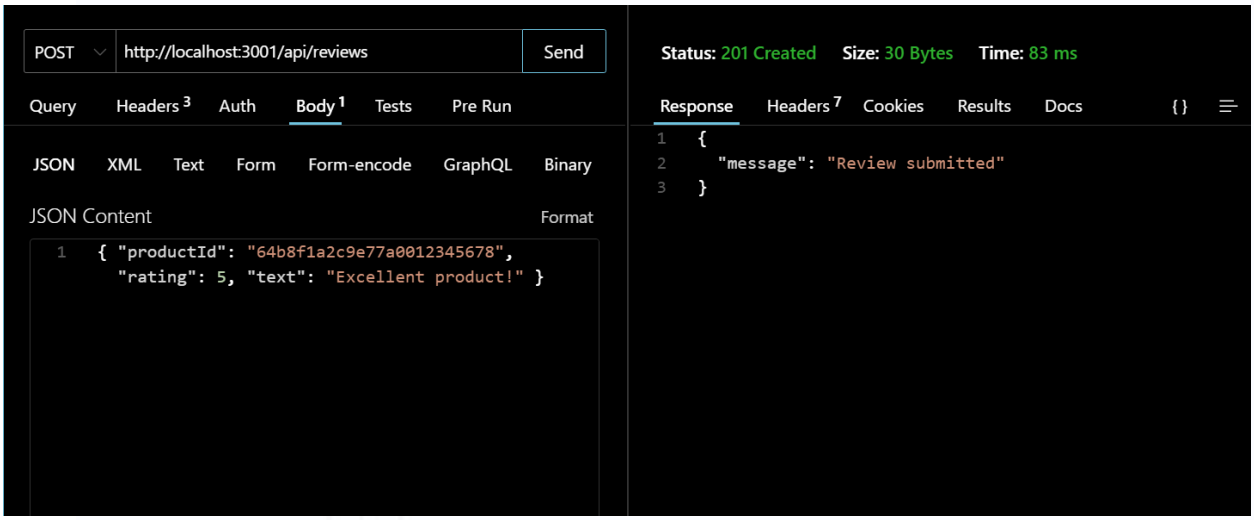
JSON Content Format

```
{
"productId": "64b8f1a2c9e77a0012345678",
"quantity": 2
}
```

Status: 200 OK Size: 26 Bytes Time: 111 ms

Response Headers 7 Cookies Results Docs { }

```
{
  "message": "Cart updated"
}
```


POST
http://localhost:3001/api/reviews
Send
Query
Headers 3
Auth
Body 1
Tests
Pre Run
JSON
XML
Text
Form
Form-encode
GraphQL
Binary
JSON Content
Format
1 { "productId": "64b8f1a2c9e77a0012345678",
"rating": 5, "text": "Excellent product!" }
Status: 201 Created
Size: 30 Bytes
Time: 83 ms
Response
Headers 7
Cookies
Results
Docs
{ }
1 {
2 "message": "Review submitted"
3 }